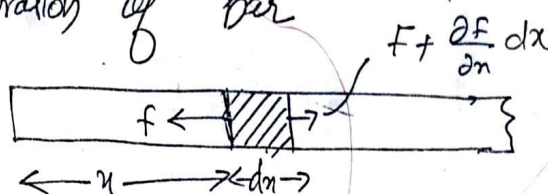


Lumped mass sys & distributed mass sys (11)

In the previous sections masses are assumed to be concentrated on a particular region. But in practical sys masses, damper, spring, constant ... etc are found to be distributed all over the entire sys. This type of sys are known as distributed mass sys.

Analysis distributed mass sys

eg: longitudinal vibration of Bar



Consider a uniform Bar for longitudinal vibration where mass is distributed along the whole body. An element dx of the BAR is considered.

force acting on the element

$$\left(F + \frac{\partial F}{\partial x} dx\right) - F = (mass) \times (acceleration \text{ of the element}) = dm \times \frac{\partial^2 y}{\partial t^2}$$

$$\frac{\partial F}{\partial x} dx = (\cancel{\rho} dx A) \left(\frac{\partial^2 y}{\partial t^2}\right) \quad \rho \rightarrow \text{mass/volume}$$

$$\frac{\partial F}{\partial x} = (\rho A) \frac{\partial^2 y}{\partial t^2}$$

Stress $\sigma = \frac{F}{A}$; also $\sigma = \epsilon E$

$$\therefore F = \epsilon E A$$

(12)

$$\frac{\partial F}{\partial x} = \frac{\partial \epsilon}{\partial x} E A$$

So

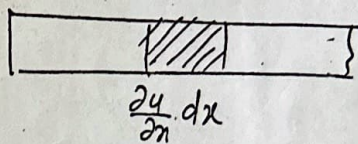
$$\frac{\partial \epsilon}{\partial x} = \rho A \left(\frac{\partial^2 y}{\partial t^2}\right)$$

consider $\epsilon = \frac{\partial y}{\partial x}$

$$\left(\frac{E}{\rho}\right) \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x}\right) = \frac{\partial^2 y}{\partial t^2}$$

$\epsilon = \frac{\text{change in length}}{\text{original length}}$

$$\left(\frac{E}{\rho}\right) \left(\frac{\partial^2 y}{\partial x^2}\right) = \frac{\partial^2 y}{\partial t^2}$$



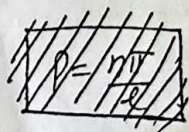
~~Wave equation~~

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{a^2} \frac{\partial^2 y}{\partial t^2} \quad \text{--- (1)}$$

→ wave eqn

where $a^2 = E/\rho$

On solving (1) we get



$$u(x,t) = \sum_{n=1}^{\infty} \left[A \sin \frac{P}{a} x + B \cos \frac{P}{a} x \right] \left[C \sin pt + D \cos pt \right]$$

where A, B, C, P are constants.

longitudinal vibration of bar

(15)

General solution of longitudinal vibration of bar is

$$u(x,t) = \sum_{n=1}^{\infty} \left[A \sin \frac{p}{a} x + B \cos \frac{p}{a} x \right] (C \sin pt + D \cos pt)$$

where $p = 2\pi f$, $a = \sqrt{\frac{E}{\rho}}$

- ① Derive the frequency equation for a free-free beam with zero initial displacement.



The general solution for longitudinal vibration

$$u(x,t) = \left[A \sin \frac{p}{a} x + B \cos \frac{p}{a} x \right] [C \sin pt + D \cos pt] \quad \text{--- (1)}$$

boundary conditions $\left(\frac{\partial u}{\partial x} \right)_{x=0} = 0$; $\left(\frac{\partial u}{\partial x} \right)_{x=l} = 0$

~~$$\frac{\partial u}{\partial x} = \left[A \cos \frac{p}{a} x \times \frac{p}{a} + (-B \sin \frac{p}{a} x \times \frac{p}{a}) \right] (C \sin pt + D \cos pt)$$~~

put $x=0$

$$\frac{\partial u}{\partial x} = \left(A \frac{p}{a} \right) (C \sin pt + D \cos pt) = 0$$

$\therefore A = 0$

Sub in (1)

$$u(x,t) = \left[B \cos \frac{p}{a} x \right] [C \sin pt + D \cos pt]$$

put $x=l$

(15)

$$\frac{\partial u}{\partial x} = \left[A \cos \frac{p}{a} l \times \frac{p}{a} + (-B \sin \frac{p}{a} l \times \frac{p}{a}) \right] [\cos pt + D \cos pt]$$

$$\frac{\partial u}{\partial x} = \left[-B \sin \frac{p}{a} l \times \frac{p}{a} \right] [\cos pt + D \cos pt] = 0$$

$$\therefore \sin \frac{p}{a} l = 0$$

$$\text{In general } \sin n\pi = 0$$

$$\text{So } n\pi = \frac{p l}{a}$$

$$p = \frac{n\pi a}{l}$$

$$a f = \frac{n\pi a}{l}$$

$$f = \frac{n}{2l} a$$

$$n \rightarrow 1, 2, 3, \dots$$

- Q Derive suitable expression for longitudinal vibrations for a rectangular uniform cross-section bar of length l fixed at one end and free at the other end. (16)

Ans ✓



$$u(x,t) = \left[A \sin \frac{p}{a} x + B \cos \frac{p}{a} x \right] (C \sin pt + D \cos pt)$$

Boundary conditions are

$$u_{x=0} = 0 \quad ; \quad B = 0$$

$$\left(\frac{\partial u}{\partial x} \right)_{x=l} = 0$$

$$\frac{\partial u}{\partial x} = \left[A \cos \frac{p}{a} x \times \frac{p}{a} + B \sin \frac{p}{a} x \times \frac{p}{a} \right] (C \sin pt + D \cos pt)$$

put $x=l$

$$0 = \left(A \cos \frac{p}{a} l \times \frac{p}{a} \right) (C \sin pt + D \cos pt)$$

$$\therefore \cos \frac{p}{a} l = 0$$

$$\text{In general } \cos \frac{n\pi}{2} = 0$$

$$\frac{p}{a} l = \frac{n\pi}{2}$$

$$p = \frac{n\pi a}{2l}$$

$$2\pi f = \frac{n\pi a}{2l} \quad ; \quad f = \frac{n a}{4l}$$

$$f = \frac{n}{4l} \sqrt{\frac{E}{\rho}}$$

$$a = \sqrt{\frac{E}{\rho}}$$

Transverse Vibration of beams

(12)

General Solution is

$$y(x,t) = A \cosh cx + B \sinh cx + C \cos cx + D \sin cx$$

- ① Derive frequency Equation for a beam with both ends free and having transverse vibration.

Ans

$$y(x,t) = A \cosh cx + B \sinh cx + C \cos cx + D \sin cx$$

$$\text{where } c^2 = \rho \sqrt{\frac{EA}{EI}}$$

$$\text{boundary conditions } \frac{d^2 y}{dx^2}(0,t) = 0, \frac{d^2 y}{dx^2}(l,t) = 0$$

$$\frac{d^3 y}{dx^3}(0,t) = 0, \frac{d^3 y}{dx^3}(l,t) = 0$$

$$\frac{dy}{dx} = A \sinh cx \cdot c + B \cosh cx \cdot c - C \sin cx \cdot c + D \cos cx \cdot c$$

$$\frac{d^2 y}{dx^2} = c^2 [A \cosh cx + B \sinh cx - C \cos cx - D \sin cx] \quad \text{--- (1)}$$

$$\frac{d^3 y}{dx^3} = c^3 [A \sinh cx + B \cosh cx + C \sin cx - D \cos cx] \quad \text{--- (2)}$$

$x=0$ in (1)

$$\frac{d^2 y}{dx^2} = c^2 (A - C) = 0 ; \quad A = C$$

$x=0$ in (2)

$$\frac{d^3 y}{dx^3} = c^3 [B - D] = 0 ;$$

$$B = D$$

(18)

 $x=l$ in ①

$$\frac{d^2 y}{dx^2} = c^2 [A (\cosh cl - \cos cl) + B (\sinh cl - \sin cl)] = 0$$

 $x=l$ in ②

$$\frac{d^2 y}{dx^2} = c^2 [A (\sinh cl + \sin cl) + B (\cosh cl - \cos cl)] = 0$$

$$A (\cosh cl - \cos cl) + B (\sinh cl - \sin cl) = 0$$

$$A (\sinh cl + \sin cl) + B (\cosh cl - \cos cl) = 0$$

Equating we get

$$(\cosh cl - \cos cl)^2 - (\sinh^2 cl - \sin^2 cl) = 0$$

$$\cosh^2 cl + \cos^2 cl - 2 \cosh cl \cos cl - \sinh^2 cl + \sin^2 cl = 0$$

we know $\boxed{\cosh^2 cl - \sinh^2 cl = 1 \text{ \& } \cos^2 cl + \sin^2 cl = 1}$

$$\text{So } -2 \cosh cl \cos cl = -2$$

$$\cosh cl \cos cl = 1$$

above equation can be solved for cl to find the natural frequency of the system.

Q Find frequency equation of a uniform beam fixed at one end and free at the other for transverse vibrations. (19)

Ans

$$y(x,t) = A \cosh cx + B \sinh cx + C \cos cx + D \sin cx$$

boundary conditions are

$$y(0,t) = 0, \quad \frac{dy}{dx}(0,t) = 0$$

$$\frac{d^2 y}{dx^2}(l,t) = 0, \quad \frac{d^3 y}{dx^3}(l,t) = 0$$

$$y(0,t) = 0 \Rightarrow 0 = A + C, \quad A = -C$$

$$\frac{dy}{dx} = C [A \sinh cx + B \cosh cx - C \sin cx + D \cos cx] \quad \text{--- (1)}$$

$x=0$ in (1)

$$0 = C [B + D]; \quad B = -D$$

$$\frac{d^2 y}{dx^2}(l,t) = C^2 [A (\cosh cl + \cos cl) + B (\sinh cl + \sin cl)] = 0$$

$$\frac{d^3 y}{dx^3}(l,t) = C^3 [A (\sinh cl - \sin cl) + B (\cosh cl + \cos cl)] = 0$$

Similarly in previous case

$$[\cosh cl + \cos cl]^2 - (\sinh^2 cl - \sin^2 cl) = 0$$

$$\cosh^2 cl + \cos^2 cl + 2 \cosh cl \cos cl - \sinh^2 cl + \sin^2 cl = 0$$

$$\text{Solving we get } \Rightarrow 2 \cosh cl \cos cl + 2 = 0$$

$$\cosh cl \cos cl + 1 = 0$$